

2. Shortest Job First (SJF)

C Program :

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, j, temp;
```

```
    int bt[20], wt[20], tat[20];
```

```
    float avg_wt = 0, avg_tat = 0;
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 0; i < n; i++) {
```

```
        printf("Enter Burst Time for P%d: ", i + 1);
```

```
        scanf("%d", &bt[i]);
```

```
    }
```

```
    // Sorting burst times in ascending order
```

```
    for (i = 0; i < n - 1; i++) {
```

```
        for (j = i + 1; j < n; j++)
```

```
        {
```

```
            if (bt[i] > bt[j])
```

```
            {
```

```
                temp = bt[i];
```

```
                bt[i] = bt[j];
```

```
                bt[j] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
    wt[0] = 0;
```

```
    for (i = 1; i < n; i++)
```

```
    {
```

```
        wt[i] = wt[i - 1] + bt[i - 1];
```

```

    }
    printf("\nProcess\tBT\tWT\tTAT\n");
    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
        avg_wt += wt[i];
        avg_tat += tat[i];
        printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);
    }
    avg_wt = avg_wt / n;
    avg_tat = avg_tat / n;
    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
    return 0;
}

```

Shell Program (sjf.sh) :

```

#!/bin/bash

echo "Enter Number of Processes"

read n

for ((i = 0; i < n; i++))
do
    echo "Enter Burst Time for P$((i + 1))"
    read bt[$i]
done

# Sorting burst times (Ascending Order)

for ((i = 0; i < n - 1; i++))
do
    for ((j = i + 1; j < n; j++))
    do

```

```

        if [ ${bt[i]} -gt ${bt[j]} ]
        then
            temp=${bt[i]}
            bt[$i]=${bt[j]}
            bt[$j]=$temp
        fi
    done
done
wt[0]=0
echo
echo -e "Process\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i = 0; i < n; i++))
do
    if [ $i -ne 0 ]
    then
        wt[$i]=$((wt[i - 1] + bt[i - 1]))
    fi
    tat[$i]=$((wt[i] + bt[i]))
    total_wt=$((total_wt + wt[i]))
    total_tat=$((total_tat + tat[i]))
    echo -e "P$(i + 1)\t${bt[i]}\t${wt[i]}\t${tat[i]}"
done
avg_wt=$(awk "BEGIN {printf \"%.2f\", $total_wt / $n}")
avg_tat=$(awk "BEGIN {printf \"%.2f\", $total_tat / $n}")

echo
echo "Average Waiting Time = $avg_wt"

```

```
echo "Average Turnaround Time = $avg_tat"
```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano sjf.sh
(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x sjf.sh
(ganesh@LAPTOP-R3RDH9NG)~$ bash sjf.sh
Enter Number of Processes
3
Enter Burst Time for P1
12
Enter Burst Time for P2
13
Enter Burst Time for P3
14

Process BT      WT      TAT
P1      12      0      12
P2      13     12     25
P3      14     25     39

Average Waiting Time = 12.33
Average Turnaround Time = 25.33
```